SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO5	**Assessment:**	Assessment Tool 1 – **Research Paper Review / Literature Survey**
Weightage:	20%	**Total Marks:**	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	y.hema sai lahari	**Reg. No.:**	192311446
Date:	31-08-26	**Duration:**	60 minutes

Code of Conduct

I y.hema sai lahari(192311446) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:y.hema sai lahari

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem.

The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.
5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.
7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

LITERATURE SURVEY REPORT

Title: Temple Donation and Management System

1. Introduction

Temple Donation and Management Systems are digital platforms designed to manage temple activities, devotee information, donations, fund allocation, events, and administrative operations. Traditional temple management often depends on manual records, paper-based receipts, spreadsheets, and separate financial records. These approaches can result in data duplication, delays, difficulty in tracking donations, and limited transparency.

Recent research has investigated digital donation platforms, blockchain-based donation tracking, financial management, and transparent fund allocation. These studies show that technologies such as blockchain, smart contracts, role-based access control, cloud systems, and web-based applications can improve transparency, security, traceability, and efficiency in donation management.

The purpose of this literature survey is to study existing approaches related to donation and fund management and identify research gaps that can be addressed through a Temple Donation and Management System.

2. Objectives of the Literature Survey

The main objectives are:

1. To study existing digital donation and fund management systems.
2. To analyze technologies used for secure donation management.
3. To compare different approaches based on transparency, security, efficiency, and scalability.
4. To identify limitations in existing systems.
5. To identify research gaps related to temple-specific donation and management.
6. To suggest possible improvements for a comprehensive Temple Donation and Management System.

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3. Individual Research Paper Reviews

Paper 1: Blockchain-Based Fund Management System for Indian Temples

Authors: Rajnandan Ray, Saurabh Vaidya, Shreya Shirbhate, Gaurav Rai, and C.V. Andhare
Year: 2025

Objective

The study aims to improve transparency and security in Indian temple fund management by using blockchain technology.

Methodology

The proposed system uses blockchain and smart contracts to record donations and automate fund allocation. It also provides role-based access for administrators and devotees.

Technologies

- Blockchain
- Ethereum
- Smart Contracts
- React.js
- Node.js
- Express.js
- MongoDB
- JWT

- MetaMask

Key Findings

The system provides tamper-resistant donation records, real-time tracking, automated fund allocation, and improved accountability. The study also identifies scalability and accessibility as areas requiring further improvement.

Limitations

- Web-based accessibility can limit users who prefer mobile applications.
- Blockchain transactions may introduce scalability and performance concerns.
- The system focuses mainly on fund management rather than complete temple administration.

Research Gap

A comprehensive system combining donation management with temple events, devotee management, receipts, reporting, and administrative functions is still required.

Paper 2: Blockchain-Based System for End-to-End Donations Monitoring

Authors: Salman Rain Ullah Mir, Yaman Ahmed Atef Kalaji, Raja Wasim Ahmad, and Atta ur Rehman Khan
Year: 2024

Objective

The research focuses on improving the tracking and tracing of donations through blockchain technology.

Methodology

The system uses blockchain and smart contracts to manage donation-related processes among registered stakeholders.

Key Findings

The research identifies centralized donation systems as having problems such as single points of failure, limited accessibility, security issues, immutability concerns, and weak data provenance. The proposed blockchain approach improves traceability and transaction management.

Limitations

- Requires blockchain infrastructure.
- Registered-party access can limit general accessibility.
- Implementation may be more complex than a traditional database system.

Research Gap

A simpler and more user-friendly system is required for temple environments where administrators and devotees may have different technical requirements.

Paper 3: Securing Charity Donation System using Blockchain

Authors: Kalaivazhi Vijayaragavan, P. S. Divya Bharathi, R. Karthika, and R. Sri Vijaya Harini
Year: 2023

Objective

The research aims to improve trust, accountability, and transparency in charitable donation management.

Methodology

The study proposes a blockchain-based donation system where transactions can be securely recorded and tracked.

Key Findings

Blockchain is identified as a useful technology for improving transparency and reducing concerns related to donation allocation and accountability.

Limitations

- The approach is primarily focused on charity donation management.
- Temple-specific functions are not addressed.
- Blockchain infrastructure may increase system complexity.

Research Gap

The approach can be extended to include temple-specific activities such as religious events, devotee registration, donation receipts, and temple administration.

Paper 4: Financial Management Practices in Islamic Donation-Based Crowdfunding Platforms

Authors: Muhammad Iqmal Hisham Kamaruddin, Nurul Aini Muhamed, Rafisah Mat Radzi, Wan Nur Fazni Wan Mohamad Nazarie, et al.
Year: 2023

Objective

The research studies financial management practices, governance, financial operations, sustainability, and financial disclosure in Islamic donation-based crowdfunding platforms.

Methodology

The researchers conducted interviews with six Islamic donation-based crowdfunding platforms and analyzed their financial management practices.

Key Findings

The research highlights the importance of governance, financial monitoring, fund management, sustainability, and financial disclosure in donation platforms.

Limitations

- The research focuses on crowdfunding platforms rather than temple management.
- It is based on a limited number of platforms.
- It does not provide a complete software architecture for temple administration.

Research Gap

There is a need for an integrated temple system that combines financial management, donation tracking, administrative control, and transparent reporting.

Paper 5: An Intelligent Blockchain-Enabled Framework for Transparent Multi-Stakeholder Donation Management

Authors: Ling-Chun Liu, Wanbing Zhan, Wei-Wei Shi, Chin-Ling Chen, et al.
Year: 2026

Objective

The research proposes a transparent donation management framework involving multiple stakeholders such as donors, organizations, financial institutions, recipients, and regulators.

Methodology

The framework uses consortium blockchain, smart contracts, role-based architecture, and automated donation workflows.

Technologies

- Consortium Blockchain
- Smart Contracts
- FISCO BCOS
- Role-based architecture

Key Findings

The prototype improves traceability, transparency, and trust while providing acceptable throughput and latency. It also supports automated governance and monitoring.

Limitations

- Multi-stakeholder blockchain architecture can be complex.
- Implementation requires specialized infrastructure.
- The framework is designed for general charitable organizations rather than temples.

Research Gap

The architecture can be adapted to temple environments by incorporating temple administrators, devotees, event management, donation categories, receipts, and financial reports.

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4. Comparative Literature Survey

Author / Year	Research Problem	Methodology	Technologies / Tools	Key Findings	Limitations	Research Gap
Ray et al., 2025	Temple fund transparency	Blockchain + Smart Contracts	Ethereum, React, Node.js, MongoDB	Improves transparency and donation tracking	Scalability and web accessibility	Complete temple management not covered
Mir et al., 2024	Donation tracking	Blockchain + Smart Contracts	Blockchain	Improves traceability and data integrity	Complex infrastructure	Temple-specific requirements not addressed
Vijayaragavan et al., 2023	Charity transparency	Blockchain	Blockchain	Improves accountability and trust	General charity focus	No temple management modules
Kamaruddin et al., 2023	Financial management	Interviews / Case Study	Donation platforms	Highlights governance and financial disclosure	Limited platform sample	Need integrated software solution
Liu et al., 2026	Multi-stakeholder donation management	Consortium Blockchain + Smart Contracts	FISCO BCOS	Improves transparency and traceability	Complex implementation	Needs adaptation for temple operations

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5. Comparative Analysis

The reviewed studies show that traditional donation systems have challenges related to transparency, accountability, security, and tracking. Blockchain-based approaches provide immutable records and improve the traceability of financial transactions.

The 2025 temple-focused study directly addresses temple fund management and demonstrates the usefulness of blockchain and smart contracts. However, its focus is primarily on financial transactions. Other studies address general charitable donations and financial governance but do not consider the complete requirements of temple administration.

Therefore, there is an opportunity to develop a more comprehensive Temple Donation and Management System that combines secure donation processing with devotee management, event management, receipt generation, donation history, fund reports, and administrative dashboards.

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6. Limitations of Existing Systems

The literature identifies the following limitations:

1. Many systems focus only on donation or fund tracking.
2. Temple-specific administrative requirements are not fully addressed.
3. Some blockchain-based systems require complex infrastructure.
4. Scalability can become an issue with increasing transactions.
5. Some systems provide limited mobile accessibility.
6. Financial reporting may not be integrated with donation management.
7. Existing systems may not provide a unified dashboard for temple administrators.
8. Donor history and digital receipt management are not always integrated.
9. Event and temple activity management are generally separate from donation systems.

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7. Identified Research Gap

Based on the literature survey, the major research gap is the lack of a **comprehensive and user-friendly Temple Donation and Management System** that integrates donation management with general temple administration.

Existing research mainly concentrates on one or two areas such as blockchain-based donation tracking, financial transparency, or fund allocation. A unified system can combine:

- Devotee registration and management

- Online donation management
- Digital receipt generation
- Donation history
- Donation category management
- Temple event management
- Fund and expense tracking
- Admin dashboard
- Financial reports
- Role-based access control
- Secure authentication
- Transparent transaction records

The research gap therefore lies in developing an integrated system that provides **security, transparency, usability, administrative efficiency, and complete temple-specific management in a single platform.**

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8. Proposed Improvements / Future Research Directions

The proposed Temple Donation and Management System can address the identified gaps through:

1. **Centralized Management Dashboard**
   Provide administrators with a single dashboard to manage devotees, donations, events, and reports.
2. **Digital Donation System**
   Allow devotees to make donations digitally and receive instant confirmation.
3. **Digital Receipts**
   Automatically generate and store donation receipts.
4. **Donation History**
   Allow devotees to view their previous donations and transaction details.
5. **Role-Based Access Control**
   Provide different access levels for super administrators, temple administrators, and devotees.
6. **Transparent Fund Tracking**
   Provide reports showing donation collection and fund utilization.

7. **Event Management**
   Allow temple administrators to create and manage religious events and activities.
8. **Secure Database**
   Use authentication, authorization, encryption, and secure database practices.
9. **Blockchain Integration as Future Enhancement**
   Blockchain can be introduced for high-value or critical financial records to provide tamper-resistant transaction history.
10. **Mobile Application**
    A future mobile application can improve accessibility for devotees.
11. **Analytics and Reporting**
    Data analytics can be used to understand donation trends and support financial planning.

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9. Conclusion

The literature survey demonstrates that digital technologies can significantly improve donation management, financial transparency, security, and accountability. Blockchain and smart contracts are particularly useful for maintaining tamper-resistant transaction records and improving traceability.

However, most existing studies focus on donation tracking, charity management, or financial transparency rather than complete temple administration. The temple-specific study demonstrates the feasibility of blockchain-based fund management but still leaves opportunities for integrating other temple activities.

Therefore, the proposed Temple Donation and Management System aims to bridge this research gap by providing an integrated platform for donation management, devotee management, digital receipts, event management, financial tracking, reporting, and secure administration. The system can initially use a conventional secure web architecture and can later incorporate blockchain, mobile applications, and analytics as future enhancements.

10. References

[1] R. Ray, S. Vaidya, S. Shirbhate, G. Rai, and C. V. Andhare, "Blockchain-Based Fund Management System for Indian Temples," *International Journal for Research in Applied Science and Engineering Technology*, vol. 13, no. 6, pp. 2073–2077, 2025, doi: 10.22214/ijraset.2025.72567.

[2] S. R. U. Mir, Y. A. A. Kalaji, R. W. Ahmad, and A. ur Rehman Khan, "Blockchain-Based System for End-to-End Donations Monitoring," in *Proc. 24th IEEE International Arab Conference on Information Technology (ACIT)*, 2023/2024, doi: 10.1109/ACIT58888.2023.10453793.

[3] K. Vijayaragavan, P. S. Divya Bharathi, R. Karthika, and R. Sri Vijaya Harini, "Securing Charity Donation System using Blockchain," *International Journal of Advanced Research in Science, Communication and Technology*, vol. 3, no. 6, 2023.

[4] M. I. H. Kamaruddin, N. A. Muhamed, R. M. Radzi, W. N. F. W. M. Nazarie, et al., "Financial management practices in Islamic donation-based crowdfunding (DCF) platforms in Malaysia," *Future Business Journal*, vol. 9, art. no. 32, 2023.

[5] L.-C. Liu, W. Zhan, W.-W. Shi, C.-L. Chen, et al., "An intelligent blockchain-enabled framework for transparent multi-stakeholder donation management," *Discover Artificial Intelligence*, vol. 6, art. no. 929, 2026, doi: 10.1007/s44163-026-01960-3.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and	Identifies relevant limitations and research gaps	Identifies basic gaps	Research gaps are unclear or missing

	identifies significant, well-supported research gaps		with limited analysis	
Future Directions & Relevance(15%)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation, Citations & References(15%)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	**Mark**
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25